A simple AXI4-Stream Video Pattern Generator

Interfaces, protocol rules and RTL design patterns

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About this guide

Accompanying source code

[**TODO**] Point at the repository layout, e.g. `m_axis.v`, `s_axis.v`, testbenches and simulation scripts.

Part I

AXI4-Stream Fundamentals

Chapter 1

Introduction to Streaming Interfaces

1.1 Why a streaming interface?

[**TODO**] Memory-mapped vs. streaming data movement; when each is appropriate.

1.2 Where AXI4-Stream sits in the AMBA family

[**TODO**] AXI4 / AXI4-Lite / AXI4-Stream comparison table.

Table 1.1: Comparison of the AMBA AXI interface flavours.

	AXI4	**AXI4-Lite**	**AXI4-Stream**
Addressing	[**TODO**]	[**TODO**]	[**TODO**]
Channels	[**TODO**]	[**TODO**]	[**TODO**]
Bursts	[**TODO**]	[**TODO**]	[**TODO**]
Typical use	[**TODO**]	[**TODO**]	[**TODO**]

1.3 Terminology: transfer, packet, frame, stream

[**TODO**] Define each level of the hierarchy and how they map to signals.

1.4 Roadmap of the document

[**TODO**] One paragraph per part.